

TOPIC 6 · AUTOMATION & AI · EXAM CASE

KICKOFF
Wed 6 May · class**SUBMISSION**
Wed 13 May · 23:59**FORMAT**
Groups of 3

How to use this document. The first two sections (*The situation at VoltRide* and *Theoretical framework*) are the theory of Topic 6. Study them: part of the rubric measures how well you apply these concepts. The following sections are the case, the rules, the deliverables and the rubric. At the end there is a glossary for quick reference.

The situation at VoltRide

VoltRide has just closed its Series B and is preparing for Series C. Before the next round, the COO needs to demonstrate to investors that operations are **scalable and mature**. As part of that plan, he launches an internal program called “**Operation Red Yarn**”: untangling the operational knots detected during the last quarter.

In **Topic 5** you analysed VoltRide’s event log with Process Mining and uncovered three uncomfortable findings: **11.3%** of production orders skip QC, international orders take **2.3×** longer than domestic ones, and the “happy path” covers only **62%** of cases. Behind those numbers there are three broken processes. Your assignment: pick one and build a bot that combines RPA + generative AI to fix it.

VoltRide takes you on as sprint consultants. You have one week, free choice of tools, synthetic data, and a single non-negotiable requirement: that your solution shows judgment. Where you put the human, where the RPA, and where the LLM, and why.

Theoretical framework

This section gives you the vocabulary and concepts you will apply. Although we are not covering this topic in a traditional lecture, the content explained here is required and is part of what the rubric evaluates. It is organised in four blocks: the evolution of the workflow, the two pieces of the stack (RPA and LLM), the integration patterns, and the risks.

1 • From the traditional workflow to hyperautomation

We call workflow the set of automated steps that move data or execute actions across systems. The first tools (classic BPM, rule engines, scripts) were rigid: if a form changed, you had to reprogram. Then iPaaS arrived (Integration Platform as a Service: Zapier, Make, n8n), which allowed connecting APIs without writing code. The next layer is RPA, which automates interaction with interfaces that have no API. And the current layer, hyperautomation, combines all of the above with AI to make decisions that previously required a human.

For your case, what matters is that the choice of tool is not technical but a matter of judgment: does the system have an API? Is the format structured? Is the decision deterministic or does it require semantic understanding? Each answer points to a different piece of the stack.

2 • RPA • what it does and what it does not

RPA (Robotic Process Automation) is the family of tools that automate repetitive, deterministic tasks: reading a file, moving an email to a folder, copying data from a screen into an Excel sheet, clicking a button. The rule of thumb: if the task can be described step by step without ambiguity (“if cell A2 contains «urgent», move to the Priority folder”), then it is a candidate for RPA.

Common tools include Power Automate Desktop (Microsoft), UiPath, Automation Anywhere, n8n (open source), Make.com, Zapier, Bizagi RPA, and for technical profiles Python with libraries such as pyautogui, Selenium, requests, or pandas. What they all share: they are quick to implement, reliable, auditable, and completely blind to the meaning of the content. RPA does not understand, it executes.

3 • LLMs • changing the rules

An LLM (Large Language Model) is a generative AI model trained on billions of texts that learns to predict the next token with surprising accuracy. What matters for you is not the internal architecture (Transformer, attention, embeddings), but what tasks it solves out of the box: text classification, entity extraction (NER), summarisation, translation, drafting, semantic comparison, reasoning over text, and — with well-designed prompts — decision-making.

Where RPA fails (interpreting a poorly written email, deciding whether an invoice is legitimate, drafting an empathetic reply), the LLM shines. Where the LLM fails (absolute precision, bit-by-bit

repeatability, error-free arithmetic), RPA shines. The combination is what we call hyperautomation, and it is the pattern shared by all three threads in this case.

4 · Risks, ethics, and regulation

Any intelligent automation project faces three risk blocks: technical (hallucinations, reliability), ethical (bias, discrimination, job displacement), and regulatory (GDPR, EU AI Act). The rubric evaluates all three.

KEY CONCEPT · Technical risk mitigation · auditability

Every production bot must record a structured log: (1) what prompt it sent to the LLM, (2) what output it received, (3) what the bot decided, (4) what action it executed. Without that trace you cannot debug errors or demonstrate diligence in an audit. Logs also let you measure hallucination rate, average processing time, and cost (each LLM call costs tokens = money).

KEY CONCEPT · EU AI Act · how it affects you today

The EU AI Act, in force since August 2024, classifies AI systems by risk level: prohibited / high / limited / minimal. A bot that classifies support emails (Thread A) is “limited risk” — it only requires minimum transparency to the user. A bot that decided whether to grant credit would be “high risk” — external audit, registration in the European database, mandatory human supervision. Your three threads fall into limited or minimal risk. In the report, justify the category.

KEY CONCEPT · GDPR applied · personal data and LLM APIs

If your bot processes personal data (emails with names, B2C customer data), you must comply with the GDPR: (1) have a legal basis for processing, (2) minimise what you send to the LLM (do not send the whole email if you only need the category), (3) consider where the LLM is hosted — Anthropic and OpenAI offer DPAs and EU-region options, others do not, (4) keep a record of processing activities. Critical point: sending PII to public APIs without a DPA contract is illegal in the EU. In this case you will use synthetic data to avoid the issue, and you will document it.

Your mission · three threads to choose from

Each group picks ONE thread from the three below. All three have the same depth and the same weight in the grade. What changes is the problem domain and which piece of the stack takes the lead. Read all three before deciding.

THREAD A	OPERATION RED YARN Intelligent triage of customer support emails
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Problem

Customer Service receives ~120 emails per day at support@voltride.com. Today one person reads them one by one, classifies them, looks up the order number in the ERP, and reassigns each email to the right department. It takes 3 hours daily, and urgent emails (battery not charging, scooter accident) sometimes get stuck behind routine commercial enquiries.

What the bot has to do

1. RPA · Watch the support@voltride.com inbox (this can be a local folder with .eml/.txt files or a test IMAP account).
2. AI · Classify each email into {Order, Warranty, Return, Sales, Other} and extract {order # if present, scooter model, urgency 1–5}.
3. RPA · Record the result in a tickets Excel/CSV/Sheets file.
4. RPA · Move the email to a folder according to the category (or tag it).
5. Human-in-the-loop · If urgency ≥ 4 , trigger a notification (Teams, Slack, email).

KEY CONCEPT · NER · Named Entity Recognition applied to Thread A

NER is the task of identifying and extracting concrete entities from free text: order numbers, models, dates, names, amounts. It used to be a specialised NLP task that required training custom models; today any LLM does it well with a well-built prompt. The “extract order # and urgency” step is exactly NER.

Starting data

You will use synthetic data. I recommend generating 25–40 emails with an LLM, validating that they make sense (subject, sender, realistic body, mix of all 5 categories). Document how you generated them.

**THREAD
B****OPERATION RED YARN**
Intelligent validation of supplier delivery notes**Problem**

PowerTech Asia (the critical supplier of batteries and motors — remember it from Topic 4?) sends its delivery notes as PDFs by email. Today an intern opens each one, looks up the matching purchase order in Odoo, and verifies that quantities and prices match line by line. It takes 4 h/day. When something does not match, it is usually detected 5 days late, with the components already on the production line.

What the bot has to do

1. RPA · Watch a /delivery_notes/ folder for new PDFs (or a webhook endpoint).
2. AI · Extract from the PDF a JSON: {po_number, date, lines: [{product, quantity, unit_price}]}.
3. RPA · Open a “purchase_orders.xlsx” file (simulating Odoo) and find the matching PO.
4. RPA · Compare quantities and prices line by line. Flag deviations > 5% or different quantities.
5. AI · If there is a deviation, draft an email to the procurement manager explaining the affected line, the delta, and the likely operational impact.
6. RPA · Save the email as a DRAFT (do not send). Record the incident in a log.

KEY CONCEPT · IDP · Intelligent Document Processing

IDP is the pattern of “read an unstructured document and return structured data”. It replaces the intern who copies data by hand. Key piece of Thread B. If the PDF is native (selectable text), a simple extractor + LLM is enough. If it is scanned, it must first go through OCR (Tesseract, Azure Form Recognizer, Google Document AI).

KEY CONCEPT · Fuzzy matching · the hidden technical challenge of Thread B

The delivery note says “Battery 48V Std”, the PO says “EL-BAT-48V-STD”. They are the same product but an exact match fails. The classical solution is fuzzy matching (Levenshtein distance, cosine similarity over embeddings). The cleanest approach for your level: ask the LLM to do the match returning a confidence score, and accept automatically only matches with score > 0.9. Below that, flag for human review.

Starting data

Generate 8–12 synthetic delivery-note PDFs (Word exported to PDF works fine) and a purchase_orders.xlsx with the corresponding POs. Deliberately introduce 2–3 deviations so the bot has something to catch.

**THREAD
C****OPERATION RED YARN**
Automatic detection of QC anomalies**Problem**

The Process Mining work in Topic 5 revealed that 11.3% of production orders are closed without going through QC-Inspection. SOP-QA-001 explicitly forbids this. Today nobody monitors it: the problem only surfaces when a customer returns a defective scooter. The COO wants a system that watches the event log and alerts in real time.

What the bot has to do

1. RPA · Load VoltRide's event log (CSV) every N minutes (or whenever a new file arrives).
2. RPA · Filter orders closed in the last 24 h and check whether they include the QC-Inspection event.
3. AI · For each non-conforming order, draft a mini-report (one paragraph) explaining which order, which product, which steps it followed, and why it breaks the SOP.
4. AI · Estimate the likely cause from the step pattern (e.g. "QC was skipped because the order went from Production directly to Packaging"; "QC was executed but without an approval stamp").
5. RPA · Email the report to the supervisor, update "qc_dashboard.xlsx", and raise a visible alert if the rate exceeds 15%.

KEY CONCEPT · Conformance Checking · the pattern behind Thread C

A core concept of Process Mining (Topic 5): detect deviations between the actual process (event log) and a reference model. Example: SOP-QA-001 states that every production order must go through QC-Inspection before reaching Done. The conformance checker compares the real log against the rule and reports the non-conforming cases. Your Thread C is a real-time conformance checker.

KEY CONCEPT · Why Thread C closes the entire course

It ties together the four pieces of MIST: process modelling (T3 · BPMN), ERP data (T4 · Odoo), discovering the anomaly with Process Mining (T5), and intelligent automation so that the anomaly stops happening (T6). It is the "from insight to action" pattern: knowing there is a problem is not enough — you have to close the loop with an automatic detection-and-response mechanism. It is exactly what a junior Business Analytics professional should be able to propose in their first job.

Starting data

Use VoltRide's event log from Topic 5. If you need it expanded or with extra cases, you can generate additional synthetic rows with an LLM. Document it.

Ground rules

Aspect	Rule
Format	Groups of 3. Individual, pair, or quartet submissions are not accepted.
Automation tool	Free choice. Power Automate Desktop, n8n (free, self-hosted), Make.com, Zapier, Python with libraries (pyautogui, requests, pandas), Node-RED, Bizagi RPA, UiPath Community, Apache Airflow — whichever fits best. The choice scores in the rubric if it is justified.
LLM	Free choice. ChatGPT, Claude, Copilot, Gemini, Perplexity, Mistral, local Llama. Whichever you prefer and can integrate.
Required components	At least one RPA component (deterministic actions: read/write files, send emails, manipulate UIs) AND at least one generative AI component (classification, NER extraction, drafting, decision).
Data	Synthetic. Generate it yourselves (manually or with an LLM) and document the process. For Thread C you may start from the Topic 5 event log.
Reuse	You may reuse templates, examples, and open-source code. Always cite them. What you cannot do is present another group's work as your own.
Generative AI for the report	Allowed but disclosed. State which parts you generated with an LLM and how you reviewed them. Plagiarising the LLM counts the same as plagiarising another author.

What you have to submit

A single ZIP per group, uploaded to Moodle on Wednesday 13 May before 23:59. Filename: T6_LastName1_LastName2_LastName3.zip. Contents:

1. PDF report of 6–10 pages: context, chosen thread, bot architecture (with diagram), tool and LLM justification, design decisions on the human / RPA / LLM split, risk treatment (GDPR, EU AI Act, hallucinations), evaluation of results with concrete metrics, lessons learned.

2. Code and/or flow: the bot file (.py, n8n .json, Power Automate .pad, etc.) plus a README with execution instructions.
3. Synthetic data used (folder /data/) and a description of how you generated it.
4. Execution logs: at least 3 real bot runs, each showing prompt + LLM output + executed action per turn.
5. Demo video (3–5 min, no fancy editing): screen sharing + voice, showing one end-to-end execution and explaining the key decisions. Upload it to YouTube as “Unlisted” or to Google Drive with viewing permission for UAM accounts.

Schedule · one week, not a day more

Date	Milestone	Detail
Wed 6 May	Kickoff class (today)	Case presentation, group formation, thread selection, and a 30-min kickoff with all three threads on the board.
Fri 8 May	Working session in class	Bring your draft architecture, the tool you have decided on, and any blockers. We will use the class to unblock you.
Sat 9 – Tue 12 May	Build phase	Implementation, synthetic data, testing, report, video. I recommend cutting scope early: a simple working bot beats an ambitious half-broken one.
Wed 13 May · 23:59	Final submission	Single ZIP on Moodle. Server time, not yours.

Rubric · how I grade you

Grade out of 10. Five blocks with explicit weight.

Block	Weight	What I evaluate
Process diagnosis	15%	“As-is” analysis: current flow, pain points, baseline metrics. “To-be” design: what the bot automates and what stays with the human. Clear justification.

Block	Weight	What I evaluate
Technical design	25%	Bot architecture (diagram). Clean split human / RPA / LLM with explicit criteria. Tool and LLM choice justified. Error handling.
Functional implementation	25%	The bot runs end-to-end in a demo. Coherent synthetic data. Real logs showing prompt + output + action. Reproducible by following the README.
Risk and regulation	15%	Explicit treatment of hallucinations (mitigation), GDPR (PII, DPA, minimisation), EU AI Act (justified risk category), auditability (logs).
Report + video	20%	Clear, concise report — no filler. Demo video that is understandable. The work product itself shows that every member contributed.

Grading bands

Band	Label	What defines it
9 – 10	Excellent	Functional bot, sound judgment, risks well covered, exemplary documentation. The report could be used as a reference in future editions.
7 – 9	Good	Functional bot with some minor limitation. Correct analysis. A risk slightly weak or some implementation detail still pending.
5 – 7	Pass	Bot that works in the basics. Analysis present but superficial. Risks treated only generically.
3 – 5	Fail (recoverable)	Incomplete bot or poor documentation. A second attempt may be useful.
0 – 3	Fail	No working bot, or the work does not demonstrate understanding of Topic 6.

Bonus · up to +0.5 on the final grade

I add up to half a point if your work shows any of these four elements without being asked: (1) serious treatment of non-obvious edge cases (illegible emails, corrupted PDFs, duplicated events), (2) multi-channel integration (email + Slack + dashboard), (3) a predictive component beyond the LLM (regression, clustering, ranking), (4) a serious justification of why this tool was chosen over the alternatives. Not stackable beyond +0.5.

FAQ · questions I have already been asked**Which tool do you recommend if I have never touched RPA?**

For non-technical profiles, n8n (self-hosted or cloud) and Make.com are the fastest way to get something working. For Python profiles, writing the bot by hand with pandas + requests + an LLM client library (anthropic, openai) is usually cleaner than fighting with a visual tool. Power Automate Desktop is ideal if you plan to do screen scraping of desktop applications.

How do I connect the LLM to my flow?

Three options depending on your stack: (a) a direct HTTP call to the provider's API (Anthropic, OpenAI), (b) a native node in n8n / Make / Zapier, (c) using the LLM as a copilot during development and leaving the final calls hardcoded with already-validated outputs (last resort, lower mark, because the bot loses adaptability).

How much time should I budget?

Plan for 12–18 hours per person spread over the week. Roughly half for understanding and design, half for implementation and documentation. If you go beyond 25 hours per person, stop and simplify: the case does not require an industrial-grade product.

What do I do if a teammate is not pulling their weight?

Talk it out among yourselves first. If it is not resolved, send me a three-way email (everyone in copy) explaining the situation before Tuesday 12 May. If I confirm the issue, that member receives a reduced individual grade. I do not want to find out at submission time — I would rather know as early as possible.

Can I reuse code from tutorials or from previous work?

Yes, citing it. What is not accepted: copying another group's bot, copying from a public repo and pretending it is yours, or generating the entire report with ChatGPT without reviewing it. Originality is not about writing every line from scratch — it is about the judgment you apply.

How do I share the video?

Upload it to YouTube as “Unlisted” (not “Private” — I would not be able to see it) and put the link in the report. Alternative: Google Drive with “anyone with the link can view”. Vimeo is also fine. What I do not want: 800 MB .mp4 files inside the ZIP.

What if I miss the deadline?

-1 point per 24 h of delay, up to 48 h. After that the submission is not accepted. Let me know in advance if there is a serious issue.

Glossary

Terms used throughout this document and that may show up in the rubric.

Term	Definition
RPA	Robotic Process Automation. Automation of repetitive, deterministic tasks via bots that interact with systems the way a person would (click, copy, paste, read file, send email).
LLM	Large Language Model. Generative AI model trained on billions of texts. Examples: GPT-4, Claude, Gemini, Llama, Mistral.
Hyperautomation	Combination of RPA + iPaaS + generative AI + analytics to automate full processes, not just isolated tasks. Term popularised by Gartner.
NER	Named Entity Recognition. NLP task that extracts entities (numbers, names, dates, amounts) from free text. Today handled directly by LLMs.
IDP	Intelligent Document Processing. Pattern for extracting structured data from unstructured documents (PDFs, scans, emails). Combines OCR + LLM.
OCR	Optical Character Recognition. Conversion of an image into machine-readable text. Required before the LLM if the document is scanned. Examples: Tesseract, Azure Form Recognizer.
Conformance Checking	Process Mining concept. Comparison between the actual flow (event log) and a reference model (SOP, BPMN) to detect deviations.
Human-in-the-loop	Design pattern in which a human reviews or approves the bot’s actions before they are executed, especially when the LLM’s confidence is low or the action is irreversible.

Term	Definition
Prompt	Text instruction sent to the LLM. Prompt quality is one of the main drivers of output quality. “Prompt engineering” is the discipline of designing them well.
Token	The minimum unit processed by an LLM. Roughly 0.75 words in English, slightly less in Spanish. APIs charge per token.
Transformer	Neural network architecture (Vaswani et al. 2017) that underpins all modern LLMs.
Hallucination	LLM output that sounds coherent and plausible but is factually incorrect. The central risk of any system built on generative AI.
DPA	Data Processing Agreement. Contract required by GDPR between the data controller and the data processor. Determines whether you may send PII to an LLM API.
EU AI Act	European regulation (in force since August 2024) that classifies AI systems by risk (prohibited / high / limited / minimal) and imposes obligations on transparency, supervision, and registration depending on the category.

One last thing

What matters is not that your bot ends up looking like a polished product, but that you show judgment: where you put the human, where the RPA, where the LLM, and why. A modest bot that knows its own limits is worth more than an ambitious one that hallucinates without you noticing.